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Validation of assessments to accurately analyze the body composition of highly trained sitting volleyball players: A pilot study

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ABSTRACT

Body composition is a fundamental component of physical fitness related to the performance of Sitting volleyball (SV) players. Also, establishing the best method for evaluating the body composition of these para-athletes would be highly necessary for this field. The purpose of this study was (1) to describe the body composition of male and female highly trained SV players, (2) to compare the values obtained from this population by two different methods and (3) to establish validity on one of these methods. Thirteen Brazilian SV national team players (five males and eight females) participated in this study. The air-displacement plethysmography (ADP) method as the criterion assessment and the skinfolds (SF) method were conducted for each player. Results showed that there were no significant differences between the values of all players, which ADP and SF measured for body fat percentage (BF%) and body density (BD) ($p > 0.05$). We found significantly different values between male and female players for BF% by SF ($p = 0.04$) and BD by SF ($p = 0.04$). A high degree of reliability was found between ADP and SF measures for BF% and BD. There were statistically significant positive correlations between BF% and BD in all values for both methods ($p < 0.01$). This pilot study suggests that considering the magnitude of space, expense, and other limitations related to the ADP method against the SF method, we recommend using the SF method, which is a valid, viable and reliable method for measuring body composition in elite SV players.

1. Introduction

Sitting volleyball (SV) is a quick, unpredictable, and popular Paralympic team sport where players must move quickly on the SV court using their upper limbs (Jeoung, 2017). The athletes who play SV are disabled in their lower limbs, and they need to have good physical performance (Ahmadi et al., 2020a,b). For this purpose, among the fitness factors, body composition is a fundamental component of physical fitness related to athletic performance (e.g. higher muscle mass and less fat mass) (Ackland et al., 2012). There are many methods for measuring body composition, which include laboratory methods, such as air-displacement plethysmography (ADP) and dual-energy X-ray absorptiometry (DXA), or field methods, such as the anthropometric method, which uses the sum of the skinfolds (SF) to measure body density (Hernández-Martínez et al., 2022; Madsen et al., 1997).

ADP and SF methods have been used in different studies (Fields et al., 2002; Lemos et al., 2016; Mojtahedi et al., 2009). The SF method has mainly been compared to the DXA method. A study found an overestimation of the body fat percentage (BF%) by the SF method compared to the DXA in athletes with spinal cord injuries (Mojtahedi et al., 2009). According to a study by Stewart and Hannan (2000), there is a good agreement in BF% between the SF and DXA methods in male athletes. In a recent study by Lemos et al. (2016), they compared the values of the BD and BF% in national team players of three Paralympic sports by ADP and SF methods. They found no significant differences between the values obtained by ADP and SF for BF% and BD. They mentioned that

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when the ADP is not possible, estimating BD and BF% by the SF is a viable alternative for track and field and swimming para-athletes. We are replicating Lemos et al. (2016) study for SV players in this study.

In spite of the SV's apparent popularity, few publications focus on SV male and female athletes compared to the other Paralympic team sport athletes (Ahmadi et al., 2019; Kolkka and Williams, 1997), particularly for body composition. In addition, using the ADP method is related to having enough laboratory space, high expense, and other limitations against the SF method. Therefore, establishing the best method between the SF and ADP methods for evaluating the body composition of SV players would be highly necessary for this field. Our study aimed (1) to describe the BF% and BD of male and female highly trained SV players, (2) to compare the values obtained from this population by ADP and SF methods and (3) to validate assessments to accurately analyze measurements obtained from this population by ADP and SF.

2. Methods

2.1. Participants

Thirteen SV players participated in this study (five men: age = 32.8 ± 4.6 years, height = 184.4 ± 8.0 cm, seat height = 148 ± 4.4 cm, body mass = 85.6 ± 20.7 kg, BMI = 24.9 ± 3.9 kg/m2; eight women: age = 30.3 ± 8.4 years, height = 172.4 ± 7.6 cm, seat height = 136.3 ± 5.8 cm, body mass = 75.8 ± 15.7 kg, BMI = 25.4 ± 4.4 kg/m2), at the time all being part of the national Brazilian SV team. Two players (15.4%) were in the minimally disabled (MD) class, and the rest of the players (84.6%) were in the disabled (D) class of the SV classification (Only people with physical impairment, e.g. amputations, poliomyelitis, cerebral palsy) (Marszalek et al., 2019). Before beginning the test protocol, all players were informed about the study's aims and signed an informed consent form. The Research Ethics Committee of the University of Campinas No. 2.623.954 approved this study. The participants' characteristics are presented in Table 1.

Table 1
Characteristics of the participants.

Variables	All players (N = 13)	Men (n = 5)	Women (n = 8)
	Mean ± SD (CI 95%)	Mean ± SD (CI 95%)	Mean ± SD (CI 95%)
Seat height (cm)	140.8 ± 7.8 (136–145.51)	148 ± 4.4 (142.5–153.5)	136.3 ± 5.8** (131.4–141.1)
Height (cm)	177 ± 9.6 (171.2–182.8)	184.4 ± 8.0 (174.4–194.4)	172.4 ± 7.6* (166–178.7)
Age (years)	31.2 ± 7.1 (26.9–35.5)	32.8 ± 4.6 (27.2–38.5)	30.3 ± 8.4 (23.2–37.3)
SV experience (years)	10.6 ± 4.4 (7.9–13.3)	13 ± 4.7 (7.2–18.8)	9.1 ± 3.8 (5.9–12.3)
Weekly practice (hours)	10.5 ± 0.9 (9.9–11)	10 ± 0.1 (10)	10.8 ± 1 (9.9–11.6)
Body mass (kg)	79.6 ± 17.7 (68.9–90.3)	85.6 ± 20.7 (59.9–111.3)	75.8 ± 15.7 (62.6–88.9)
BMI (kg/m^2)	25.2 ± 4.1 (22.8–27.7)	24.9 ± 3.9 (20.1–29.8)	25.4 ± 4.4 (21.7–29.1)
Fat by SF (%)	25.95 ± 9.02 (20.5–31.39)	19.78 ± 8.34 (9.4–30.1)	29.81 ± 7.45* (23.5–36.1)
Fat by ADP (%)	22.85 ± 9.57 (17.07–28.64)	18.66 ± 10.14 (6.1–31.3)	25.48 ± 8.82 (18.1–32.8)
BD by SF (kg/L)	1.05 ± 0.02 (1.03–1.06)	1.06 ± 0.02 (1.03–1.08)	1.04 ± 0.02* (1.02–1.05)
BD by ADP (kg/L)	1.05 ± 0.02 (1.03–1.06)	1.06 ± 0.02 (1.02–1.08)	1.04 ± 0.02 (1.02–1.05)

Data are presented with mean, standard deviation, and 95% confidence interval.
* Difference between men and women players (P < 0.05).
** Difference between men and women players (P < 0.01).
Abbreviations: **SV**: sitting volleyball; **BMI**: body mass index; **SF**: skinfold; **ADP**: air displacement plethysmography; **BD**: body density.

2.2. Procedures

All the players were instructed to have a normal diet the day before the test. On the test day, they were hydrated, had not eaten food for 3 h after breakfast, urinated before analysis, did not use diuretic medications, wore light clothing (swimsuit or bikini), and removed all metal objects before the tests. The tests were completed within a time difference of one day and different locations (First day: ADP test in UNICAMP, Campinas, SP; and second day: SF test in SESI, Suzano, SP).

The body mass (kg) was measured using a digital scale, model Balmak® electronic Classe III, with an accuracy of 0.01 kg. A SANNY® (Personal Caprice Portable, Brazil) stadiometer with 0.1 cm precision was used for height and seat height. Body mass index (BMI) was calculated by the ratio of body mass to the square of height.

SF thickness of the chest, abdominal, front thigh, suprailiac, subscapular, triceps, and mid-axillary were measured by calipers (Lange®, UK). All measures were performed 1 cm above the selected location with the thumb and index fingers. All measurements were performed on the right side of the players' body, except when the right limbs were the amputated limbs. No athletes had both limbs' sides amputated. SF evaluations were measured according to the Jackson and Pollock (1985) procedures. The measurements were performed three times for each player by a single trained technician. The mean of the three measurements was used as its representative value for each SF. After the SF measurements, BF% was calculated using the Siri (1993) equation from the BD estimated by the seven SF protocols proposed by Jackson and Pollock (1978) for male players and by Jackson et al. (1980) for female players.

The ADP test was repeated twice, and if it was observed a significant difference (body volume>0.150 L), a third measure was performed. Also, the volume of thoracic gas for all players was measured according to the Lemos et al. (2016) method. The body composition analysis was performed using a BOD POD (Body Composition Tracking System; Cosmed, Chicago, USA). BOD POD calculates the player's BD and then uses the Siri' formula (Siri, 1993) to calculate their BF%. The players were instructed to enter the BOD POD chamber. The players were then asked to sit, limit their movements, and breathe normally throughout the ADP test (Fig. 1).

2.3. Statistical analyses

All statistical analyses were performed with IBM SPSS Statistics version 23.0. The normality of distribution was assessed by the Shapiro–Wilk test (Heirani and Ahmadi, 2012), and the results indicated a normality distribution. All variables are summarised as the means ± standard deviation and confidence interval at 95% (CI 95%). Pearson correlation coefficients and multivariate analysis of variance (MANOVA) followed by Bonferroni's post hoc tests were used to compare the BD and BF% determined by ADP and SF between both groups. Bland–Altman plots were used to illustrate the differences, biases, and agreement between SF and ADP (Bland and Altman, 1986). Intraclass correlation (ICC) was used to evaluate the reliability of each measurement method. Analyses were considered statistically significant at $p < 0.05$.

3. Results

There were no significant differences between the values of the players measured by the ADP and SF. We found significant differences between male and female SV players for BF% by SF ($p = 0.04$) and BD by SF ($p = 0.04$).

Figs. 2 and 3 show no differences between BF% (A: $r = 0.13$; $p = 0.6$) and BD (B: $r = 0.12$; $p = 0.7$). Mean differences were −3.09 ± 4.29 (−11.51–5.31) and 0.001 ± 0.01 (−0.02–0.02) for BF% and BD, respectively. For all players, the analysis of BF% in comparing ADP to SF resulted in a bias score of −3.09 ± 4.29 (%) ($t(12) = -2.59$, $p = 0.02$),

Fig. 1. Body composition test by the Bod Pod in both male and female SV players.
* The ADP evaluation test obtained without footwear.

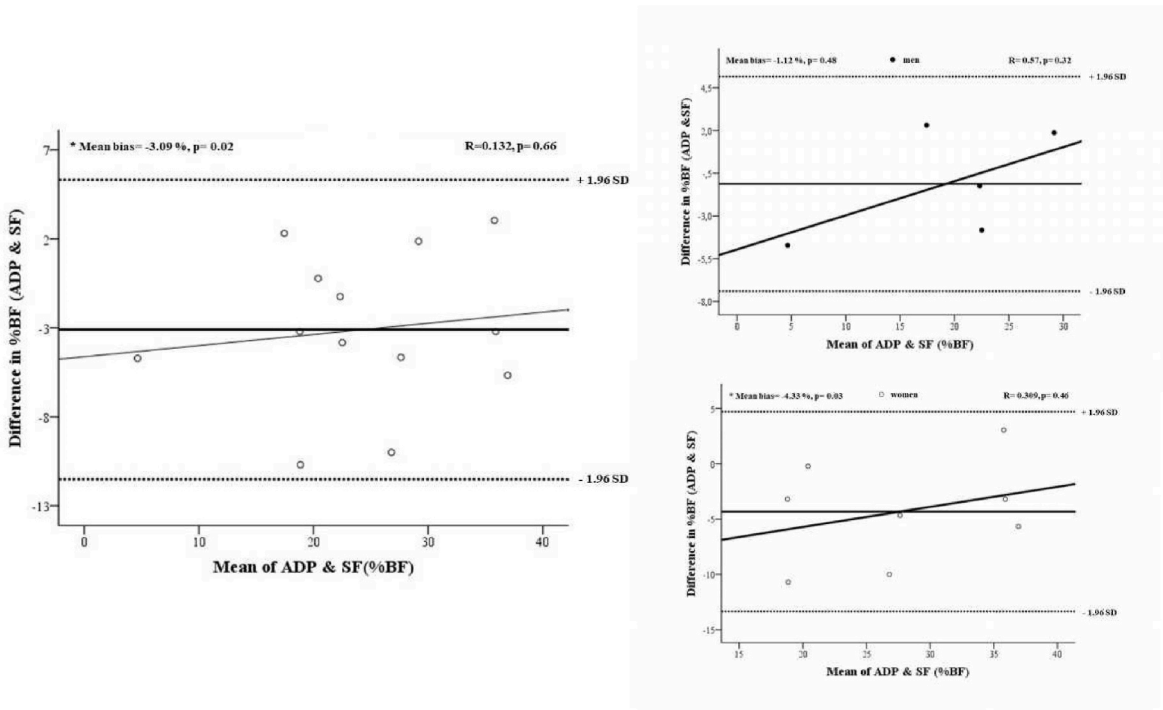

Fig. 2. Bland and Altman plots comparing body fat percentage determined by ADP and SF techniques for the male and female players and separated by gender (• men and ◦ women). Dashed black lines define the 95% limits of agreement between methods. Continuous black lines define the mean bias. Inclined and continuous black lines represent the relationship between mean and difference between SF and ADP methods.
Abbreviations: **SD**: standard deviation; **BF**: body fat; **SF**: skinfold; **ADP**: Air displacement plethysmography.

with limits of agreement between −11.51 and 5.32 (%) and no significant systematic bias (R = 0.132, SEE = 4.44) (Fig. 2), and the analysis of BD resulted in a bias score of 0.001 ± 0.01 (kg/l) (t (12) = 0.37, p = 0.71), with limits of agreement between −0.019 and 0.022 (kg/l) and no significant systematic bias (R = 0.119, SEE = 0.01) (Fig. 3). For the male players, the analysis of BF% resulted in a bias score of −1.12 ± 3.2 (%) (t (4) = 1.2, p = 0.48), with limits of agreement between −7.39 and 5.15 (%) and no significant systematic bias (R = 0.57, SEE = 3.04) (Fig. 2, • men), and the analysis of BD resulted in a bias score of −0.005 ± 0.01 (kg/l) (t (4) = 1.003, p = 0.21), with limits of agreement between 0.01 and −0.021 (kg/l) and no significant systematic bias (R = 0.501, SEE = 0.008) (Fig. 3, • men). For the female players, the analysis of BF% resulted in a biased score of −4.33 ± 4.6 (%) (t (7) = 0.79, p = 0.03), with limits of agreement between −13.35 and 4.69 (%) and no significant systematic bias (R = 0.309, SEE = 4.73) (Fig. 2, ◦ women), and the analysis of BD for the female players comparing ADP to SF resulted in a biased score of 0.005 ± 0.01 (kg/l) (t (7) = 0.016, p = 0.19), with limits of agreement between 0.025 and −0.015 (kg/l) and no significant systematic bias (R = 0.007, SEE = 0.01) (Fig. 3, ◦ women).

A high degree of reliability was found between the ADP and SF measures of BF% and BD. For all players, in BF%, the average measure ICC was 0.921 with a 95% confidence interval from 0.644 to 0.978 (F

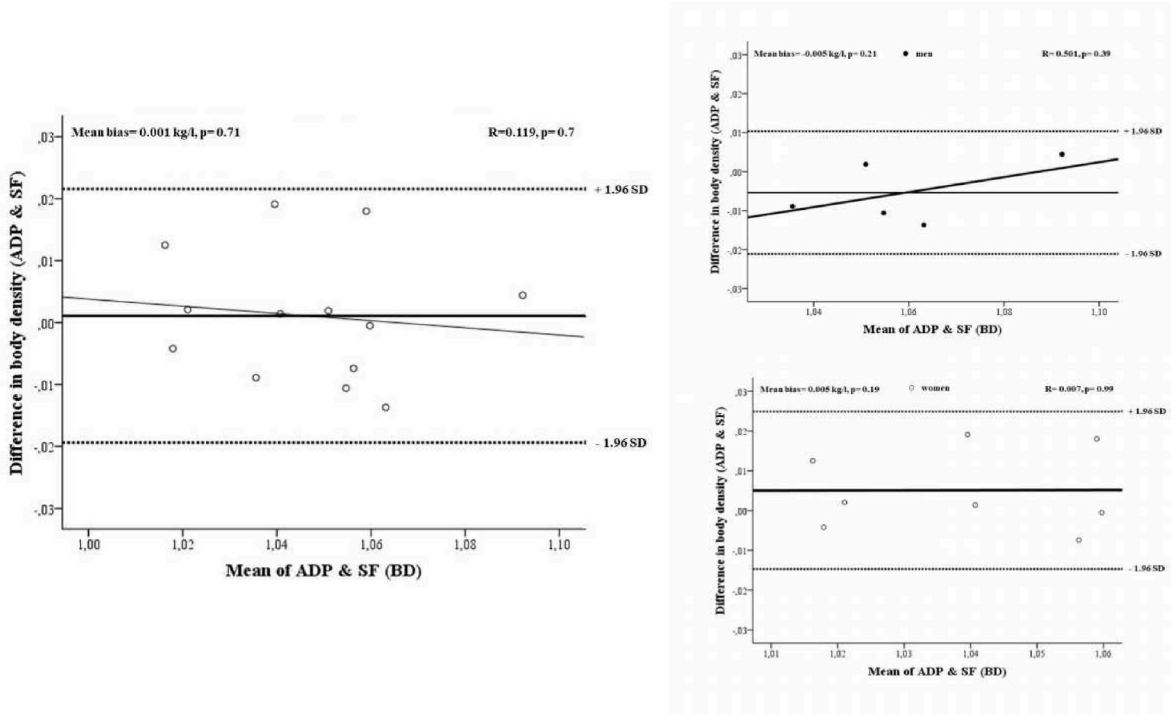

Fig. 3. Bland and Altman plots comparing body density determined by ADP and SF techniques for all players and separated by gender (• men and ◦ women). Dashed black lines define the 95% limits of agreement between methods. Continuous black lines define the mean bias. Inclined and continuous black lines represent the relationship between mean and difference between SF and ADP methods.
Abbreviations: **SD**: standard deviation; **SF**: skinfold; **ADP**: air displacement plethysmography.

(12,12) = 17.769, $p \leq 0.001$), and in BD, the average measure ICC was 0.944 with a 95% confidence interval from 0.816 to 0.983 (F (12,12) = 16.667, $p \leq 0.001$). For male players, in BF%, the average measure ICC was 0.972 with a 95% confidence interval from 0.780 to 0.997 (F (4,4) = 32.617, p = 0.003), and in BD, the average measure ICC was 0.955 with a 95% confidence interval from 0.638 to 0.995 (F (4,4) = 27.217, p = 0.004). For female players, in BF%, the average measure ICC was 0.857 with a 95% confidence interval from 0.128 to 0.973 (F (7,7) = 11.571, p = 0.002), and in BD, the average measure ICC was 0.917 with a 95% confidence interval from 0.621 to 0.983 (F (7,7) = 13.546, p = 0.001).

There were statistically significant correlations between BF% of SF and ADP in all players (r = 0.895, $p < 0.01$). For male players' the correlation between BF% of SF and ADP was r = 0.959 ($p < 0.01$), and for female players was r = 0.853 ($p < 0.01$). Also, there were statistically significant correlations between BD (kg/l) of SF and ADP in all players (r = 0.888, $p < 0.01$). For male players' the correlation between BD (kg/l) of SF and ADP was r = 0.945 ($p < 0.01$), and for female players was r = 0.863 ($p < 0.01$) (see Table 2).

4. Discussion

The purpose of the present study was firstly to describe the BF% and BD of highly trained male and female SV players and secondly to validate assessments to accurately analyze measurements obtained of this population by ADP and SF. To the best of our knowledge, this is the first study to investigate the body composition in SV players, particularly by ADP. As the first result of this study, two different non-invasive methods (ADP and SF) for measuring body composition presented similar results. Additionally, we found that the SF anthropometric method is valid for measuring body composition in SV players.

ADP and SF measurements in our study showed similar results for the BF% and BD in all players. This result was in agreement with the study of Lemos et al. (2016) for Paralympic athletes. However, Vescovi et al. (2002) stated that the %BF estimates by SF may be more accurate than those obtained by ADP, regardless of the body composition. Also, we found significant differences between the male and female SV players for the BF% and BD, which SF measured, but there were not the same results for the ADP. Lemos et al. (2016) found the same results for Paralympic Goalball and Swimming para-athletes.

The position of the players in the BOD POD chamber during the analysis could affect the parameters measured, not to a large extent (Peeters, 2012). Although our study followed the instructions of the BOD POD manufacturer concerning participant position, to avoid any possible effects, we controlled the position of all players in the BOD POD. Besides, as our participants had substantial lower limb muscle atrophy or amputation, it can affect the constitution of the fat-free mass and the BD reflecting values under or overestimated body fat (Moore

Table 2
Pearson correlation coefficients (r) between measures of body fat percentage and body density by SF and ADP.

	All players (n = 13)		Men (n = 5)		Women (n = 8)	
	BF by ADP (%)	BD by ADP (kg/L)	BF by ADP (%)	BD by ADP (kg/L)	BF by ADP (%)	BD by ADP (kg/L)
BF by SF (%)	0.895**	–	0.959**	–	0.853**	–
BD by SF (kg/L)	–	0.888**	–	0.945**	–	0.863**

**Correlation is significant at the 0.01 level.
Abbreviations: **BD**: body density; **BF**: body fat; **SF**: skinfold; **ADP**: Air displacement plethysmography.

et al., 2015). Conversely, some studies found that a different level of spinal cord injury does not seem to influence body composition, and the BD method does not differ between subjects with or without spinal cord injury (George et al., 1988). According to that, the results of the present study cannot be related to different levels of the lesion.

When comparing the differences and potential biases between the ADP and SF results, it was found that the SF overestimated BF% compared to the ADP. Similarly, Lemos et al. (2016) found that the results in Paralympic players revealed that the SF overestimating BF% (men: 24.22 ± 7.92; women: 32.35 ± 6.14) compared with the ADP (men: 22.99 ± 7.70; women: 30.32 ± 7.29). However, a study presented that SF (5 different equations) underestimates BF% compared with ADP among wheelchair basketball and wheelchair rugby athletes (Goosey-Tolfrey et al., 2016). Also, we found that BD was very similar between both methods. This result agreed with the study of Lemos et al. (2016) for Paralympic Goalball and Track athletes. Besides, Vescovi et al. (2002) found that the BD of female non-disabled college athletes was very similar between SF and ADP results. The results of our study represented that SF is a valid method for evaluating body composition in elite SV players. This finding is in agreement with Lemos et al. (2016) study.

The main limitation of the present study was the sample size of the study population. Only 13 tested elite SV players were relatively small. However, this number was the extent of the available male and female Brazilian national SV teams at the time of the study. Therefore, there was not the opportunity to test more elite SV players. On the other hand, it is necessary to continue this study on larger sample size groups of SV players at an international level to confirm the results of the present study. Additionally, we suggest that future studies use the DXA or hydrostatic weighing as gold standard reference measures of the body composition to compare with ADP and SF methods in this population.

5. Conclusion

This study showed that the ADP and SF methods have similar results for measuring the body composition in highly trained SV players. From a clinical implication perspective, the present study can be seen as a first step in establishing a reference base concerning the body composition values of these para-athletes. Besides, considering the magnitude of space, expense, and other limitations related to the ADP method against the SF method, we recommend the use of the inexpensive and easily applied SF method as a valid, viable and reliable method for measuring the body composition in elite SV players, which it helps SV coaches to monitor the training progress of the athletes.

CRediT authorship contribution statement

Shirko Ahmadi: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Project administration, Resources, Software, Supervision, Validation, Visualization, Writing – original draft, Writing – review & editing. **Lauren J. Lieberman:** Writing – review & editing. **Kwok Ng:** Writing – review & editing. **Ronaldo G. Oliveira:** Investigation, Visualization. **Gustavo L. Gutierrez:** Investigation, Visualization, Writing – review & editing. **Marco C. Uchida:** Data curation, Investigation, Supervision, Visualization, Writing – original draft, Writing – review & editing.

Declaration of competing interest

There are no known conflicts of interest associated with this paper.

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